SACSS
Topical Worksheet 2026

CANDIDATE NAME									
CENTRE NUMBER	S					INDEX NUMBER			

SCIENCE (PHYSICS) 5086

Topic: Refraction of Light **Topical Worksheet 2026**

1 hour 30 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on the work you hand in.
You may use an HB pencil for any diagrams, graphs or rough working.
Write in dark blue or black pen.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
DO **NOT** WRITE ON ANY BARCODES.

Section A (10 marks)
Answer **all** questions.
For each question choose one answer, A, B, C or D, and write its letter in the answer grid.

Section B (61 marks)
Answer **all** questions. Each question starts on a new page.
Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **17** printed pages and 1 blank page.

Section A

Answer all the questions. For each question there are four possible answers, A, B, C and D. Choose the one you consider correct and write its letter in the answer grid on page 4.

1 A ray of light in air strikes the surface of a glass block along the normal.

What happens to the ray as it enters the glass?

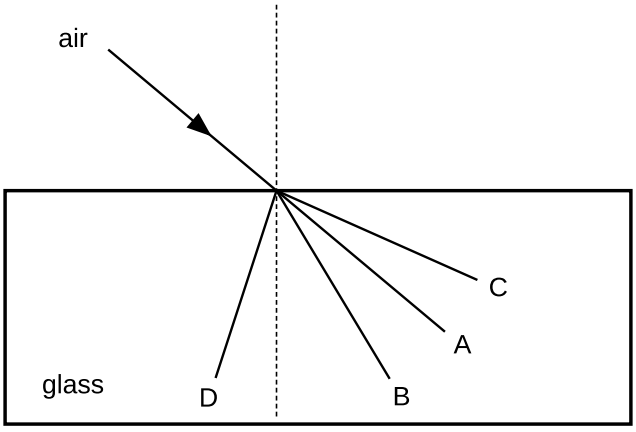
A It bends away from the normal.

B It bends towards the normal.

C It continues in the same direction.

D It is reflected back along the normal.

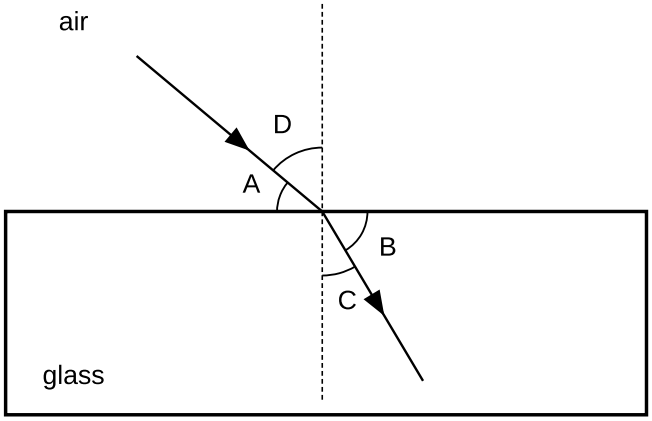
2 The diagram shows a ray of light in air striking the surface of a glass block. Four possible paths A, B, C and D are drawn inside the glass.



Which path does the ray take in the glass?

A **B** **C** **D**

3 The diagram shows a ray of light passing from air into glass. Four angles A, B, C and D are marked.



Which angle is the angle of incidence?

A **B** **C** **D**

4 Which statement about the refractive index of a material is correct?

A It has the unit m / s.

B It has no unit.

C It is less than 1 for glass.

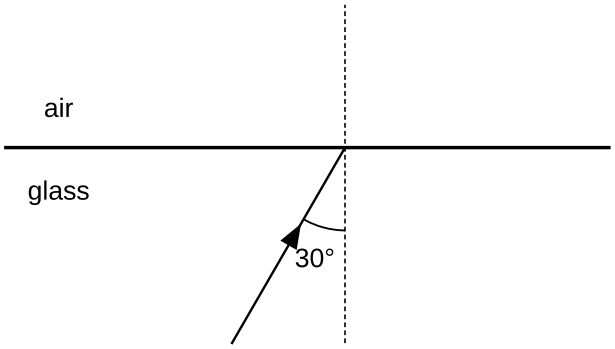
D It depends on the angle of incidence.

5 The table gives the speed of light in four transparent materials.

Which material has the largest refractive index?

	material	speed of light / 10^8 m / s
A	crown glass	1.97
B	Perspex	2.01
C	water	2.25
D	ice	2.29

6 The diagram shows a ray of light in glass striking the boundary between the glass and air. The angle of incidence is 30°.



What happens to the ray as it passes into the air?

A It bends away from the normal.

B It bends towards the normal.

C It continues in the same direction.

D It travels along the boundary.

7 A ray of light passes through a rectangular glass block. It enters one face and leaves through the opposite, parallel face.

Which statement about the ray that leaves the block is correct?

A It is parallel to the normal.

B It is parallel to the incident ray but displaced sideways.

C It is perpendicular to the incident ray.

D It travels along the same straight line as the incident ray.

8 A ray of light travels from air into glass along a path XY.

The direction of the light is reversed so that it travels from Y towards X.

What happens to the light?

A It bends further away from the normal than before.

B It travels along the normal.

C It travels back along the same path.

D It is reflected at the surface of the glass.

9 Which statement describes refraction of light?

A the bouncing back of light from a surface

B the spreading of light as it passes through a narrow gap

C the change in direction of light as it passes from one transparent material into another

D the absorption of light by a dark surface

10 Light of frequency 5.0×10^{14} Hz travels in water at a speed of 2.25×10^{8} m / s.

What is the wavelength of the light in the water?

A 2.2×10^{6} m **B** 1.1×10^{-6} m **C** 6.0×10^{-7} m **D** 4.5×10^{-7} m

Answer grid for Section A

1	2	3	4	5	6	7	8	9	10

Section B

Answer **all** the questions in the spaces provided.

11 Fig. 11.1 shows a ray of light in air striking a rectangular glass block at P. The ray travels through the block to Q. The grid has squares of side 1.0 cm.

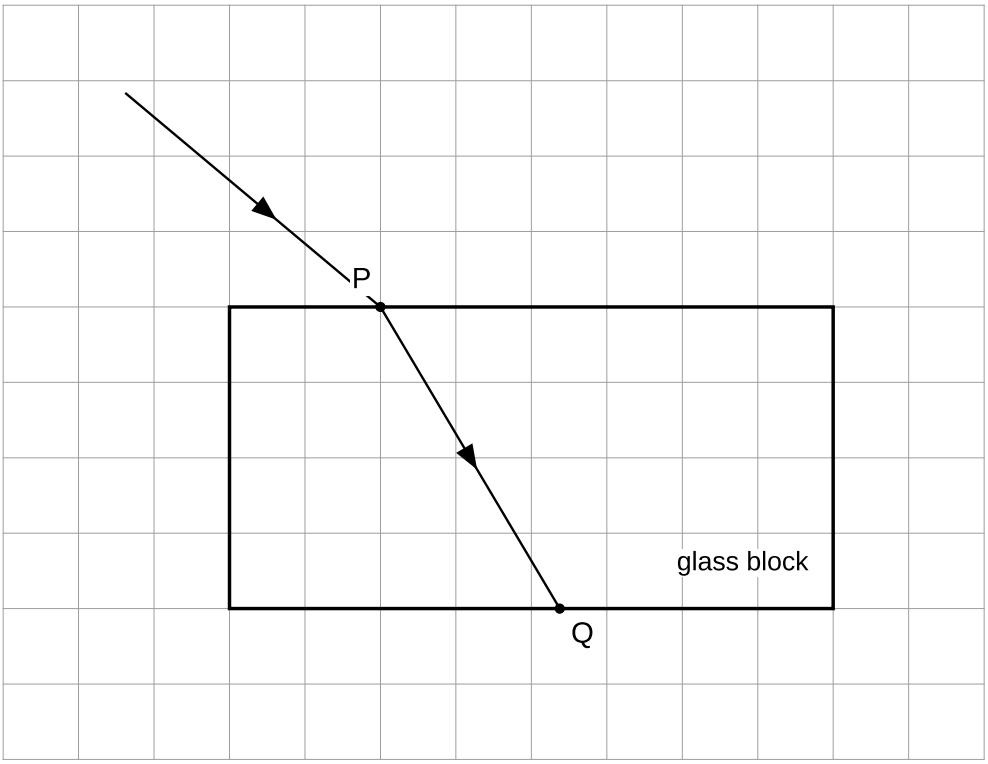


Fig. 11.1

(a) On Fig. 11.1, draw the normal at P.

Measure the angle of incidence at P.

angle of incidence = ...° [2]

(b) Measure the angle of refraction at P.

angle of refraction = ...° [1]

(c) Explain, using your measurements, why the ray bends towards the normal at P.

.....

..... [2]

(d) On Fig. 11.1, draw the normal at Q and the ray that leaves the block at Q.

[2]

[Total: 7]

12 Fig. 12.1 shows a ray of light travelling towards the centre O of the flat face of a semicircular glass block. The refractive index of the glass is 1.50. The grid has squares of side 1.0 cm.

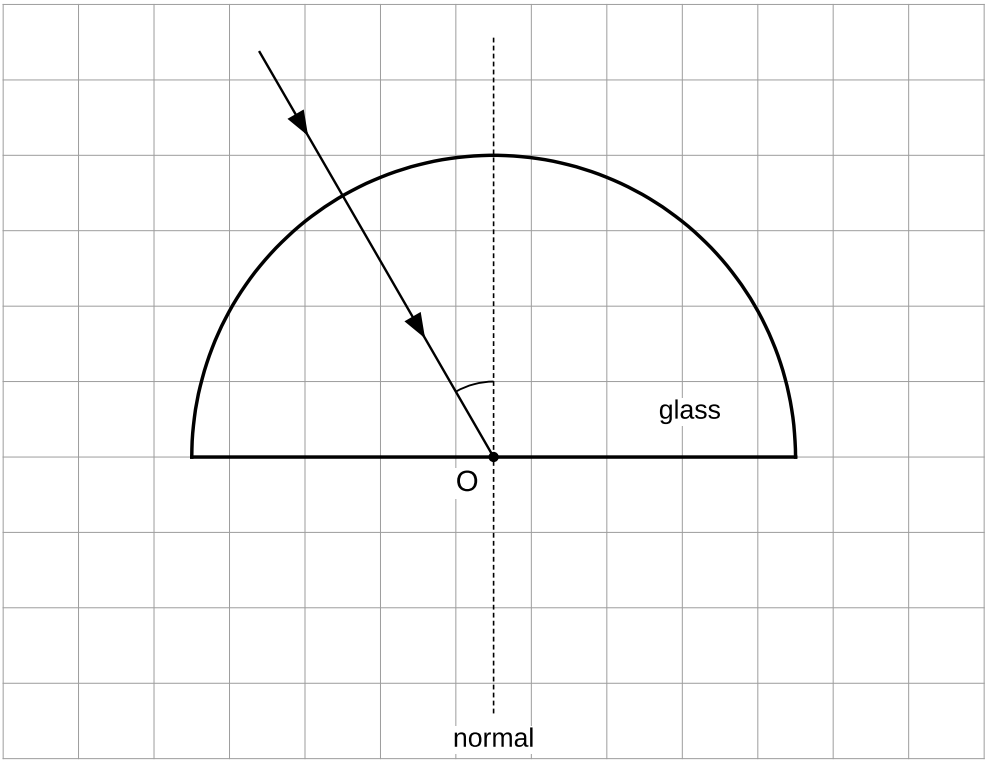


Fig. 12.1

(a) Explain why the ray does not change direction when it enters the curved surface of the block.

..... [1]

(b) The angle of incidence at O is 30°.

Calculate the angle of refraction as the ray leaves the block at O.

angle of refraction = ...° [2]

(c) On Fig. 12.1, draw the refracted ray and the weak reflected ray at O.

[2]

(d) The block is replaced by one of the same shape made of glass with a greater refractive index. The angle of incidence at O is still 30°.

State the effect on the angle of refraction at O.

..... [1]

[Total: 6]

13 A coin lies on the bottom of a swimming pool. A lifeguard lies on the edge of the pool and looks straight down at the coin. The lifeguard is not drawn to scale. Fig. 13.1 shows two rays of light from the coin that leave the water and enter her eye. On the grid, 1 square represents 0.50 m.

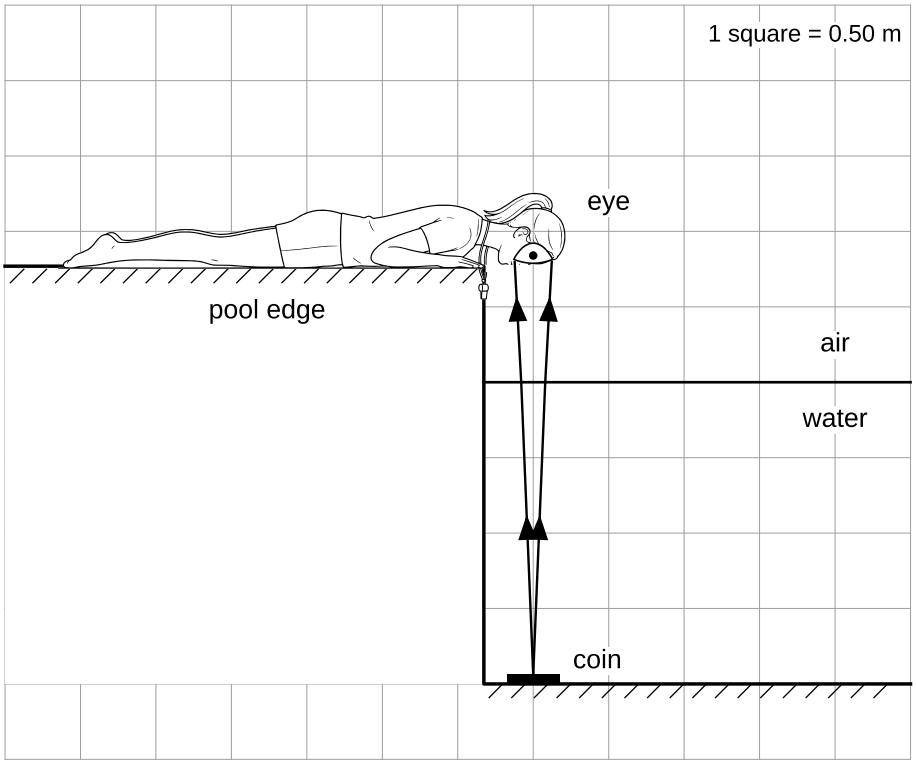


Fig. 13.1

(a) On Fig. 13.1, extend the two rays in the air backwards to locate the image of the coin. Label the image I.

[2]

(b) Use Fig. 13.1 to determine the apparent depth of the coin.

apparent depth = ...m [1]

(c) Explain why the coin appears to be nearer to the surface than it really is.

.....

..... [2]

(d) The pool is filled to the same depth with a liquid of greater refractive index than water.

State the effect on the apparent depth of the coin.

..... [1]

[Total: 6]

14 A student uses optical pins to trace the path of a ray of light through a glass block. Fig. 14.1 shows the block on a sheet of paper. Pins P_1 and P_2 are on a line that meets the block at X. The ray leaves the block at Y.

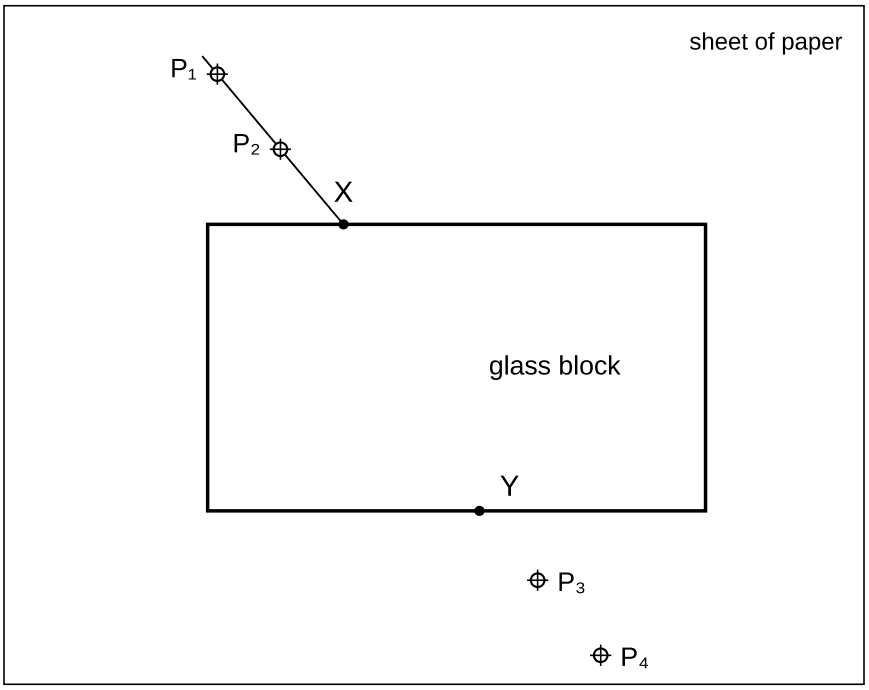


Fig. 14.1

(a) Describe how the student uses pins P_3 and P_4 to find the path of the ray that leaves the block.

.....

..... [2]

(b) On Fig. 14.1, draw the path of the ray inside the block and the normal at X. Label the angle of incidence i and the angle of refraction r.

[2]

(c) The student repeats the experiment for different angles of incidence. Table 14.1 shows the results.

angle of incidence i / °	20	30	40	50	60
angle of refraction r / °	13.2	19.5	25.4	30.7	35.3

Table 14.1

Use the results in Table 14.1 to determine the refractive index of the glass.

refractive index = ... [2]

(d) Suggest **one** precaution that makes the results more accurate.

..... [1]

[Total: 7]

15 A public aquarium has a large window made of acrylic plastic. Light travels through the acrylic at a speed of 2.0×10^8 m / s. The speed of light in air is 3.0×10^8 m / s.

(a) Calculate the refractive index of the acrylic.

refractive index = ... [2]

(b) The refractive index of the water in the aquarium is 1.33.

Calculate the speed of light in the water.

speed = ...m / s [2]

(c) The acrylic window is 0.12 m thick.

Calculate the time taken for light to travel straight through the window.

time = ...s [2]

[Total: 6]

16 A swimming pool has a red laser light show. The laser emits light of frequency 4.7×10^{14} Hz. In air, the wavelength of the light is 6.4×10^{-7} m.

(a) Calculate the speed of the light in air.

speed = ...m / s [2]

(b) The light enters the water of the pool.

State the frequency of the light in the water.

frequency = ...Hz [1]

(c) The speed of the light in the water is 2.25×10^{8} m / s.

Calculate the wavelength of the light in the water.

wavelength = ...m [2]

(d) State what happens to the speed of the light as it enters the water.

..... [1]

[Total: 6]

17 At a hawker centre, a girl looks at a straight straw standing in a glass of water. Fig. 17.1 shows the straw, the end T of the straw and her eye.

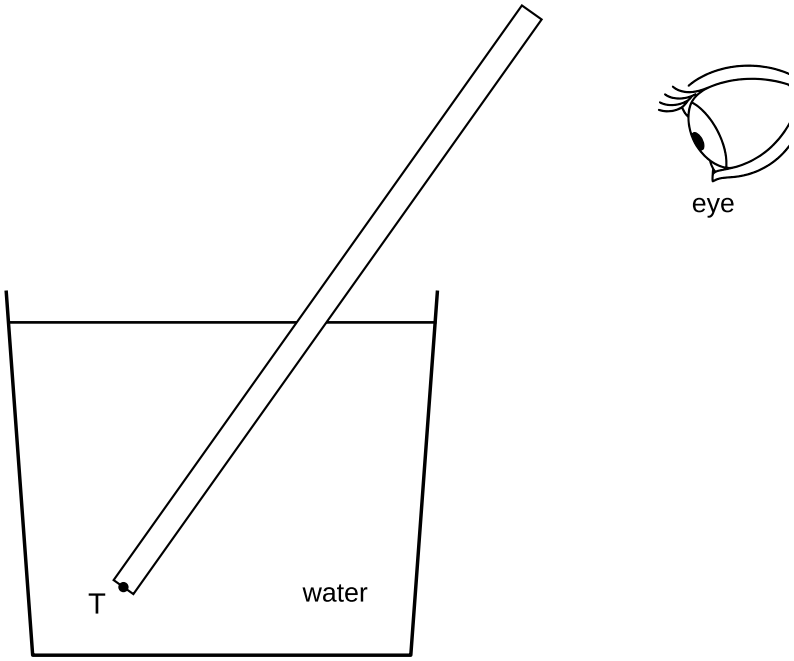


Fig. 17.1

(a) On Fig. 17.1, draw a ray of light from T that passes through the surface of the water and enters the eye. Draw the normal where the ray leaves the water.

[2]

(b) On Fig. 17.1, mark with the letter I the position where the girl sees the end T, and draw the part of the straw under the water as she sees it.

[2]

(c) Explain why the straw appears to be bent at the surface of the water.

.....

..... [2]

[Total: 6]

18 In a science show, a layer of cooking oil floats on water in a tall glass tank. Fig. 18.1 shows a ray of light in air striking the surface of the oil at P. The refractive index of the oil is 1.47 and the refractive index of water is 1.33.

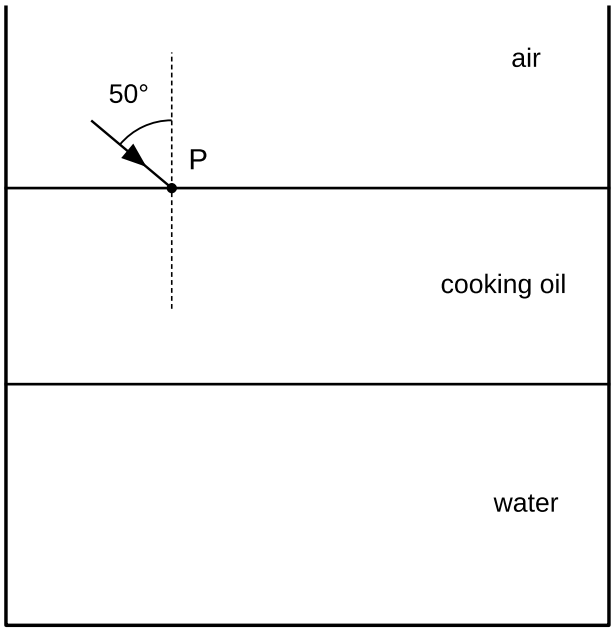


Fig. 18.1

(a) On Fig. 18.1, draw the normal at the boundary between the oil and the water and the path of the ray through the oil and through the water.

[2]

(b) The water is replaced by a liquid that has the same refractive index as the oil.

State and explain what happens to the ray at the boundary between the oil and this liquid.

.....

.....

..... [3]

[Total: 5]

19 Fig. 19.1 shows a ray of light striking face P of an equilateral glass prism. The grid has squares of side 1.0 cm.

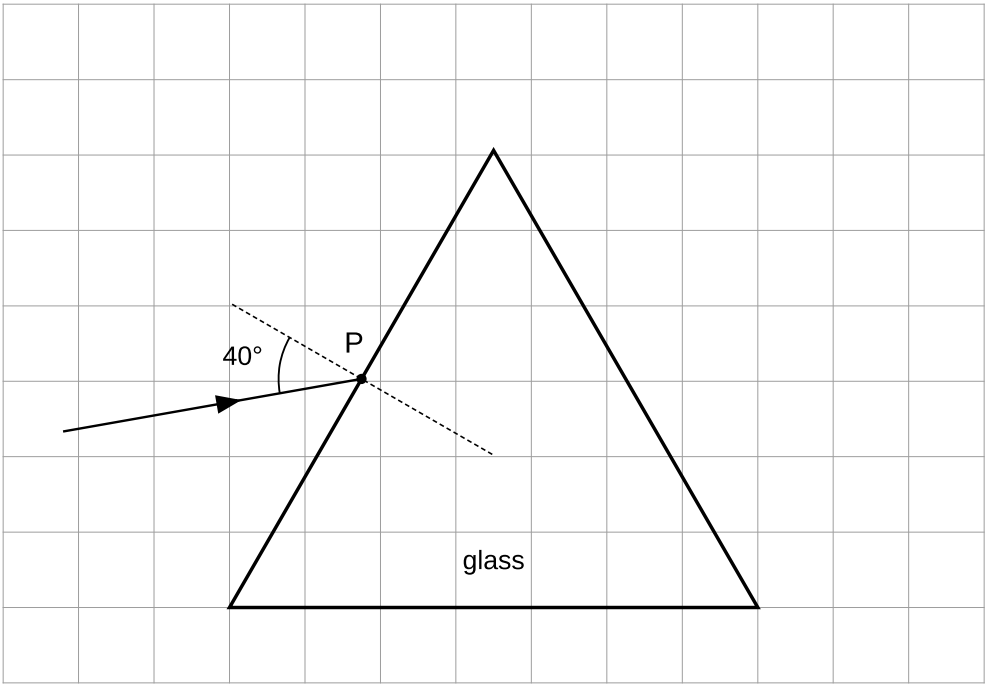


Fig. 19.1

(a) The angle of refraction at P is 25°.

On Fig. 19.1, draw the refracted ray inside the prism until it reaches the second face.

[2]

(b) Show that the angle of incidence at the second face is about 35°.

.....

..... [2]

(c) The angle of refraction as the ray leaves the second face is 61°.

On Fig. 19.1, draw the normal at the second face and the ray that leaves the prism.

[2]

[Total: 6]

20 A girl stands on the bank of a pond at night. She shines a torch at the surface of the water. Fig. 20.1 shows the ray of light from the torch striking the surface at S. A fish F is in the pond.

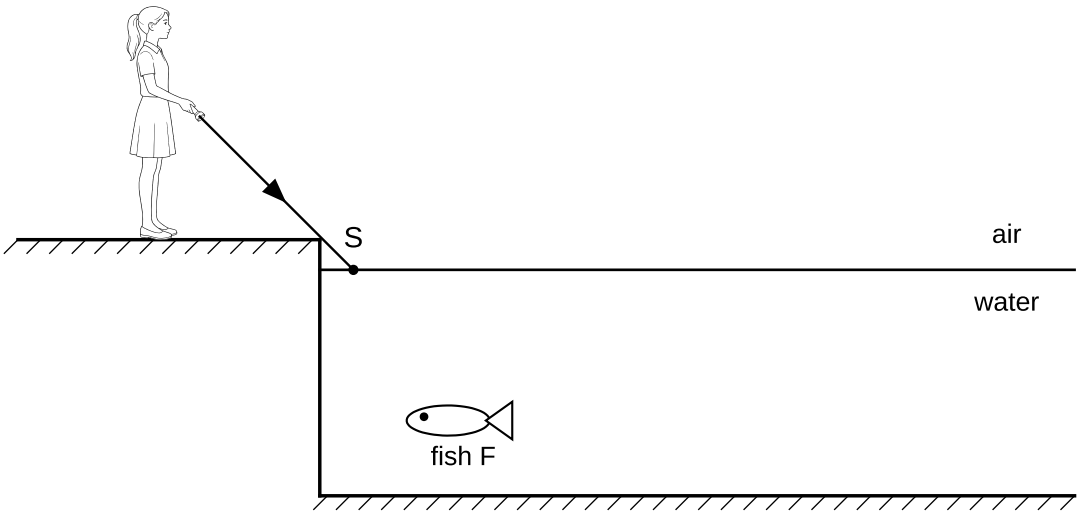


Fig. 20.1

(a) On Fig. 20.1, draw the normal at S.

[1]

(b) The angle of refraction in the water is 32°.

On Fig. 20.1, draw the refracted ray. State whether the light reaches the fish.

[1]

..... [1]

(c) Some of the light is reflected at S.

On Fig. 20.1, draw the reflected ray. State the angle of reflection.

[1]

angle of reflection = ...° [1]

(d) The girl moves the torch so that the angle of incidence at S decreases.

State what happens to the angle of refraction.

..... [1]

[Total: 6]

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